RCModalBass

User manual

RCJacH · JSFX instrument

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1 Introduction

RCModalBass is an experiment in modal string synthesis, presented as a playable electric bass. String vibration, playing gestures, pickups, electronics, and noises are purely synthesized, meaning that no recorded samples are used.

The instrument maintains up to six persistent strings. Their tuning, construction (winding, gauge, stiffness, etc.), playing position, and technique all contribute to the final sound.

The model uses one plane of string motion and simplified contact and friction. It does not represent a particular commercial bass.

2 Installation

2.1 ReaPack

1. Install ReaPack in REAPER if needed.
2. Import the RCJacH repository in ReaPack's repository manager: https://github.com/RCJacH/ReaScripts/raw/master/index.xml.
3. Synchronize repositories, find **RCModalBass**, select it for installation, and apply the changes. This will install all accompanying modules.

2.2 Manual installation

1. In REAPER, open the resource directory using **Options → Show REAPER resource path in explorer/finder**.
2. Create a folder for `RCModalBass` under `Effects`.
3. Copy `RCModalBass.jsfx` and the entire `modules` folder into that folder, preserving their relative paths:

```
RCModalBass/
  RCModalBass.jsfx
  modules/
    ... .jsfx-inc files
```

4. Refresh the FX browser.
5. When updating, replace the main file and modules together.

3 Reading this manual

LH means left or fretting hand; **RH** means right or picking hand. These labels describe musical roles regardless of which hand you use on a physical instrument.

In control names such as `String N gauge` and `Pickup N sensitivity`, `N` stands for the string or pickup number.

Monospace identifies interface labels and slider values, such as `Auto` or `0`. Bold identifies MIDI inputs and MIDI data values, such as **CC64** or **64–127**, and short labels. Italics mark concepts where they are introduced or explained, and publication titles. Technique and articulation names are capitalized in prose; words describing physical fingers, picks, or movements remain lowercase.

4 Finding your way around

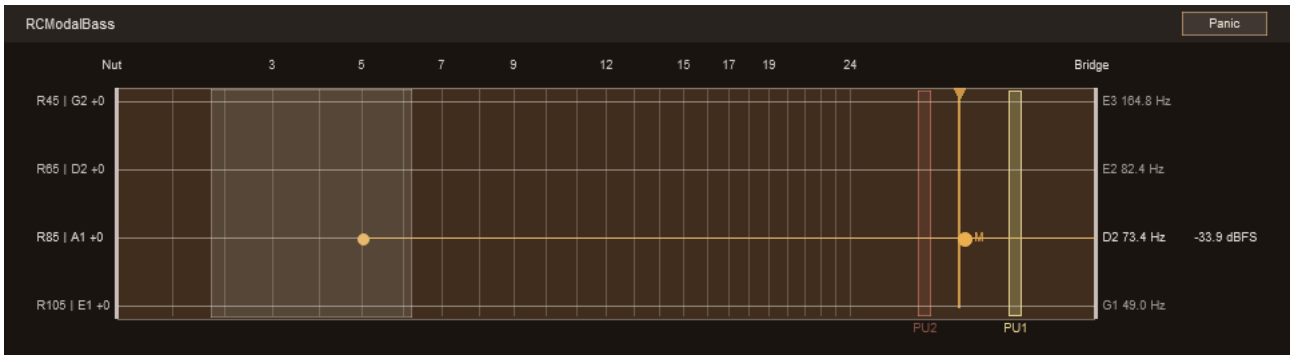
The native `Page` selector groups the controls. Some pages have subpages or show controls only for the selected model.

Page	Purpose
`Play`	Technique, hand positions, muting, vibrato, and other controls that require active manipulation.
`Setup`	MIDI routing, bend/slide ranges and timing, and settings typically left unchanged for the current piece or passage.
`Excitation`	Attack character and technique output levels.
`Instrument`	String configuration, including tuning, construction, sustain, and action.
`Pickups`	Pickup geometry, wiring, sensitivity, and electrical properties.
`Electronics`	Filtering and circuit loading.
`Fretting`	LH attack, pressure, ghost damping, and contact.
`Noise`	Buzz/clank balance, slide friction, and pick scrape settings.
`Variation`	Seeded differences between playing gestures.
`Mapping`	MIDI note and controller assignments.
`Diagnostics`	Mode count and `MIDI Reset`.

The fretboard shows hand positions, articulations, pickup placement, frequency, and string activity.

`I/M` index/middle finger
`U/D` pick up/down
`S/P` Slap/Pop
`H.O./P.O.` Hammer-on/Pull-off
`T` Tapping
`X` Ghost Note
Diamond Harmonics touch location
`P.M.` palm muting

The title-bar `Panic` button immediately silences the instrument and clears MIDI note tracking. See the MIDI reference for the gentler `Stop All Notes` gesture.



5 Playing RCModalBass

The `Play` page controls note allocation and gestures during a performance. `Setup` establishes their MIDI routing, movement limits, and response. `Output` on `Play` adjusts the final instrument level.

5.1 Notes and strings

An ordinary *playing note* is assigned to one of the enabled strings. Each string supports one ordinary note at a time and can retain vibration after key release, depending on the release gesture. A new note on that string reuses its vibration state rather than creating an independent voice.

A *fret position* sets the stopping point and therefore the *speaking length* of the string. Position zero is the *open string*; higher positions shorten the speaking length and raise pitch. The same pitch may be reachable on several strings, so choosing a string also chooses the construction and speaking length producing it.

String numbers keep their identity after retuning. Four strings use 5–2; a fifth adds low B, and a sixth adds high C. With one to three strings, the enabled slots are E, E/A, and E/A/D.

String	Default open pitch	Sounding MIDI	Default gauge
6 / B	B0	**23**	130
5 / E	E1	**28**	105
4 / A	A1	**33**	85
3 / D	D2	**38**	65
2 / G	G2	**43**	45
1 / C	C3	**48**	32

Gauge is in thousandths of an inch: 105 means .105 inches.

> **Playing notes sound one octave below incoming MIDI pitch.** Tuning controls specify sounding pitch; keyswitches and forwarded MIDI retain their literal numbers. For example, playing note **40** produces open E1, sounding note **28**.

5.1.1 Picking position

`Picking hand position` sets the requested right-hand attack midpoint, from the bridge at `0` to the configured maximum distance at `1`. Moving it toward the bridge generally gives a brighter, leaner attack. The actual attack centers also include the selected technique's spacing.

Position is captured on each new right-hand attack; moving it during sustain or Legato Slide does not reshape existing vibration. RH marks the requested midpoint, while attack dots show the captured centers. The maximum distance is configured under Performance setup.

5.2 String and fret selection

With `String selection` set to `Auto`, the instrument chooses a playable string. `Auto fingering strategy` on the `Play` page determines the preferred position:

- `Low Position`: favors lower frets.
- `Easy`: balances hand movement, span, and string changes.
- `Nearest`: favors positions close to the current hand position.
- `Lock LH Position`: keeps the hand at the selected position and limits automatic fingering to what is reachable there.

`Fretting hand position` sets the position of the fretting hand's index finger.

The automatic route also respects the hand reach and lower-fret limit configured under Performance setup.

`Open-string preference` favors nearby fretted notes, low opens, high opens, or all opens. The selections are `Fretted first`, `Low open`, `High open`, and `All open`. Low and high are determined by tuning. Preferred open drones are protected while held or sustained by `Let Ring`.

Select a string manually or hold Force String for a specific route. The pitch must fit an enabled string. Force String operates when `MIDI channel mode` is `Single channel`; string-channel modes choose the string through the incoming channel.

5.3 Playing techniques

`Playing style` selects the *excitation*, the gesture that sets the string in motion, used by incoming playing notes. Attack-character settings are covered under Shaping the attack.

5.3.1 Style and stroke selection

`Playing style` groups the excitations into *Finger*, *Pick*, Slap/Pop, and *Tapping* modes.

Style *keyswitch* velocity selects the switching behavior:

- **1–99**: *hold* temporarily; release to restore the previous applicable style.
- **100–127**: *latch* the selection after release.

A style assigned to its own CC latches at **64–127**; **0–63** rearms it. Styles sharing a CC divide its range in the displayed order; a style is selected only when the value enters a different band.

The `Force Index / Slap / Up input` and `Force Middle / Pop / Down input` assignments provide momentary stroke overrides, called **Stroke 1** and **Stroke 2** in this manual. Their effect depends on the current style; the latest held stroke wins.

5.3.2 Finger and Pick

`Finger Auto` alternates the middle and index finger models, starting with middle. `Pick Auto` alternates down and up strokes, starting down. `Finger Index`, `Finger Middle`, `Pick Up`, and `Pick Down` select fixed attacks; the stroke inputs can override them temporarily.

Both alternation sequences restart after two seconds without a right-hand attack. Alternation is shared across strings and channels; slides, HO/PO, and taps do not advance it.

5.3.3 Slap and Pop

Slap strikes inward with the thumb; *Pop* pulls outward and releases.

`Slap/Pop Auto` starts with Slap after a rest. A Slap attack establishes a split: that string, its immediate higher neighbor, and all lower strings use the Slap excitation; higher strings use the Pop excitation. For example, slapping string 5 (the E string) places the split between string 4 (the A string) and string 3 (the D string). If the next note is played on string 3 or a higher-pitched string, it triggers Pop; otherwise, it triggers Slap. Finger/Pick playing, MIDI reset, and string-count changes clear the split.

5.3.4 Tapping

The `Tapping` style can start a silent string and prefers idle strings when `String selection` is `Auto`. A lower target on the same string triggers a Pull-off. Direct open-string taps are rejected, but Pull-offs may return to open.

5.4 Articulations and phrasing

The *articulation* inputs modify how a note begins or continues. Legato Slide, Hammer-on/Pull-off, Harmonics, and Ghost Note inputs are momentary at every nonzero velocity. The newest held articulation wins; releasing it restores another held one. Their CCs use nonzero values for on and **0** for off. Send switches before the playing notes.

5.4.1 Hammer-ons and Pull-offs

Hold the default keyswitch, **Note 17 (F-1)**, and overlap reachable same-string notes. Higher targets *hammer on*; lower targets *pull off*. Keep a lower key held for a trill return: releasing the current note returns to the most recent eligible held key.

5.4.2 Legato Slide

Hold the default keyswitch, **Note 19 (G-1)**, and overlap a reachable destination note. The held string slides without replucking; unavailable routes fall back to normal allocation. Releasing the destination before arrival ends the slide.

In fretted mode, pitch steps between fret crowns while friction follows continuous movement. Fretless mode permits fractional positions.

5.4.3 Ghost Notes

Hold the default keyswitch, **Note 30 (F#0)**, for a *Ghost Note*: a muted fretted attack. Open candidates are rerouted or rejected. Releasing the switch does not restore a Ghost Note's sustain; a fresh ordinary pluck clears its mute.

5.4.4 Harmonics

Hold the default keyswitch, **Note 29 (F0)**. Switch velocity selects a *partial*; playing-note velocity controls attack strength. CC assignments use the same ranges:

Switch velocity / CC value	Requested partial	Convenient center
1–7	Ordinary-note override	—
8–23	2	**16**
24–39	3	**32**
40–55	4	**48**
56–71	5	**64**
72–87	6	**80**
88–103	7	**96**
104–127	8	**112**

0 releases the input. Each note retains its captured partial. The touch node is chosen near the LH span; automatic hand following makes the smallest needed shift, while `Lock LH Position` keeps the hand fixed.

5.4.5 Palm damping

By default, **CC67** controls the palm damping amount. Position and width define a bridge-relative patch that changes both tone and sustain, including on open and fretless strings.

5.4.6 Note endings and Let Ring

`Note-off hand gesture` chooses the release:

- `Mute held`: damp at the stopped position.
- `Lift`: release toward open and let the tail ring.
- `Lift and mute`: release toward open and damp.

`Lift` releases the string without adding an impact.

`Let Ring` (default **CC64**) defers key-release damping: **64–127** holds, **0–63** releases. Releasing the pedal applies the selected release gesture to deferred notes that have not been replaced on their strings. String reuse and `Panic` take precedence. The slider displays MIDI state.

5.4.7 Chords and overlapping notes

`Chord Mode` set to `Off` allows string reuse and releases older fretted notes made unreachable by a hand shift. `On` protects held notes, prefers free strings, and rejects impossible additions. `Hand span` still applies; opens, Tapping, and explicit string/channel choices have their own exceptions.

Changes to `Chord Mode` affect the allocation of subsequent notes.

5.5 Bend, slide, vibrato, and pressure

These inputs change held notes. Their movement ranges and onset timing are configured under Performance setup.

5.5.1 Bend and slide

Bend raises tension at a fixed fret; *slide* moves the stopping position. By default, moving the **Pitch Wheel** above center bends up; moving it below center slides down.

When bend and slide share a source, the split is at wheel value **8192** or CC/pressure value **64**. With separate sources:

Source	Bend	Slide
Wheel	Upper half: up.	Both directions around center.
CC	Full range: up.	Both directions around **64**.
Aftertouch	Full range: up.	Full range: down.

Slides stay within the configured fret range.

5.5.2 Vibrato

Vibrato periodically raises tension at a fixed stopped position. It does not simply apply an LFO to the pitch. Depth is measured in cents and specifies the maximum upward pitch excursion. By default, **CC1** scales depth after its first input; **CC19** controls rate.

Open strings do not use fretting vibrato.

5.5.3 Fretting pressure

By default, **channel aftertouch** adds pressure above a normal grip on held, stopped strings. **0** returns to normal. CC/aftertouch are continuous; note assignments are momentary at every velocity.

The tension and contact response to pressure are configured under Fretting.

5.6 Handling gestures

5.6.1 String scratching

String scratching noise occurs when the hand moves quickly along a wound string.

1. Hold Handling Noise (default **Note 14 (D-1)**).
2. Move `LH position input` or use the LH fret-key range.

Movement speed drives the texture rate and spectral character. The keyswitch velocity or CC value scales its level.

5.6.2 Right-hand landing

Right-hand landing noise occurs when an exciter (finger or pick) in the plucking hand touches and mutes a string with enough force to produce a low-frequency bump that the pickups detect.

1. Select a string manually or with Force String. Auto uses the last played string.
2. Hold Handling Noise, then press Stroke 1 or 2 (**Note 25 (C#1)** / **Note 27 (D#1)**) to trigger the landing variant for that stroke.

3. Stroke velocity sets impact strength. Keep it held for local damping; release Stroke or Handling Noise to lift silently.

A stroke already held before Handling Noise is enabled creates no new impact. Landings preserve fret/note ownership and do not advance alternation.

5.6.3 Pick scraping

Pick scraping noise is caused by sliding a pick along a wound string.

1. Select a string manually or hold Force String. Auto without a selection is silent.
2. Move `Pick scrape position` (default **CC20**).
3. First values, selection changes, and pauses establish a silent baseline; movement produces the scrape.

5.7 Performance setup

The `Setup` page determines how the instrument receives and responds to a part: which channels control it, how far the hands and pitch can move, and how note strength and timing are interpreted.

5.7.1 MIDI routing

With `MIDI channel mode` set to `Single channel`, playing expression is shared. `Ch1=6(B)-Ch6=1(C)` and `Ch1=1(C)-Ch6=6(B)` assign channels 1–6 to enabled strings in descending or ascending string-number order, skipping disabled slots. Only as many channels as there are enabled strings are assigned. With four strings, the 6-to-1 order maps E/A/D/G to channels 1–4; the reverse maps G/D/A/E. Retuning preserves this order.

Setting `Keyswitch MIDI channel` to `Follow channel mode` shares gestures in `Single channel` mode and resolves styles, articulations, and strokes independently in string-channel modes. Selecting `All` or a fixed keyswitch channel instead broadcasts those gestures.

Bend, slide, and `Let Ring` follow string-channel ownership. Continuous fretting pressure follows playing channels independently of the keyswitch filter. Instrument knobs and RH alternation remain global. `Playing style` and `Chord Mode` show the last addressed channel; manual edits supply the common base.

5.7.2 Hand reach and position

`Hand span` sets how far automatic fingering can reach from the anchor set by `Fretting hand position`. `Capo Position` sets the minimum fret considered by automatic routing, while `Number of frets` on `Instrument` sets the upper limit. Explicit string choices retain their own routing exceptions.

`Maximum excitation distance from bridge` calibrates the range of `Picking hand position`. Increasing this distance lets the same position input place the attack farther from the bridge.

5.7.3 Response and timing

`Velocity-to-displacement curve` converts playing-note velocity into excitation strength. Increasing it makes intermediate velocities produce weaker attacks relative to maximum velocity. The physical attack parameters that use this strength are described under Shaping the attack.

`Repluck damping` removes some existing motion before an ordinary Finger excitation, changing how strongly the new attack takes over the string. `Note-off damping T60` controls how quickly the string decays when the selected `Note-off hand gesture` mutes it. Shorter times produce a more abrupt stop; longer times leave a longer tail.

`MIDI bend-up range` sets the maximum pitch rise from the bend input, while `MIDI slide range` sets how far the slide input can move pitch by changing the stopping position.

`Legato slide time` sets the time taken to travel one semitone between notes, so wider intervals take longer. Increasing it slows the slide. Destination velocity and the held slide input amount also shape that travel.

`Vibrato delay` sets how long a note sounds before vibrato begins. `Vibrato build-up` then brings it gradually to the requested depth. Build-up is measured in vibrato cycles, so its duration also depends on the vibrato rate. Plucks and taps restart this onset; Legato Slide and HO/PO continue it.

5.7.4 MIDI forwarding

`MIDI passthrough` independently forwards playing notes, keyswitches, and controllers, or combinations of them, to downstream effects or instruments in the FX chain.

Forwarded notes retain their incoming pitch. The internal playing-pitch octave offset therefore does not transpose notes sent to another instrument.

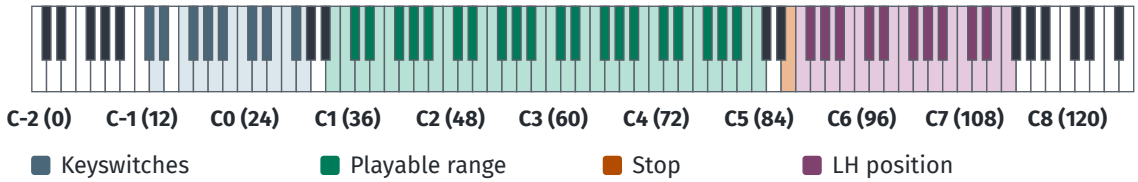
6 MIDI reference

The tables below list the default assignments on `Mapping`. The following rules describe shared mappings, controller updates, note pairing, and stopping. Channel routing and forwarding choices are described under Performance setup.

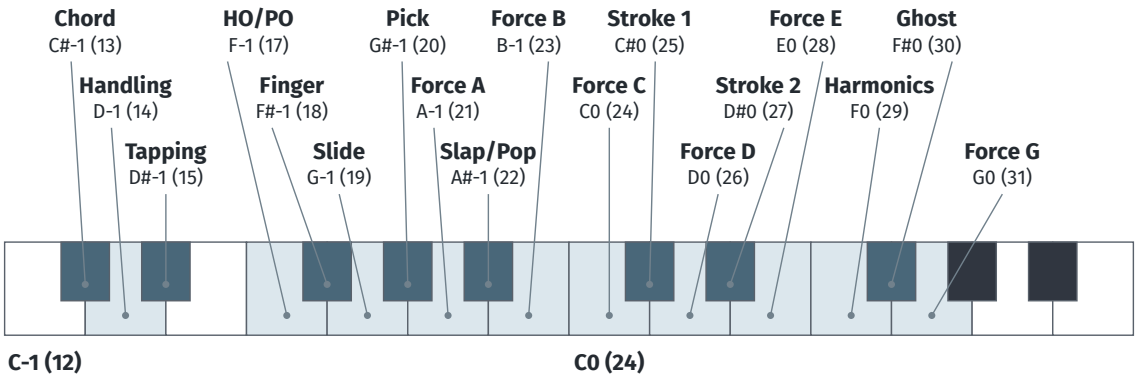
6.1 Default notes

Use these assignments when placing keyswitches in a MIDI item. Numbers are literal incoming MIDI notes; the playing-pitch octave offset does not apply to them.

All diagrams use incoming MIDI pitch names, with **C3 = note 60**. The overview shows the full MIDI keyboard; the detail panels use larger scales.

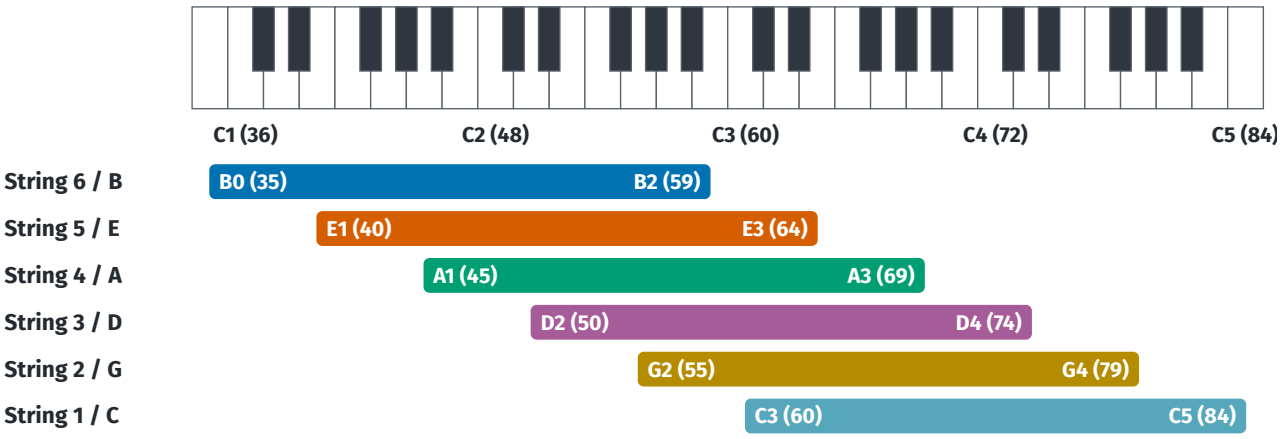


Keyswitches



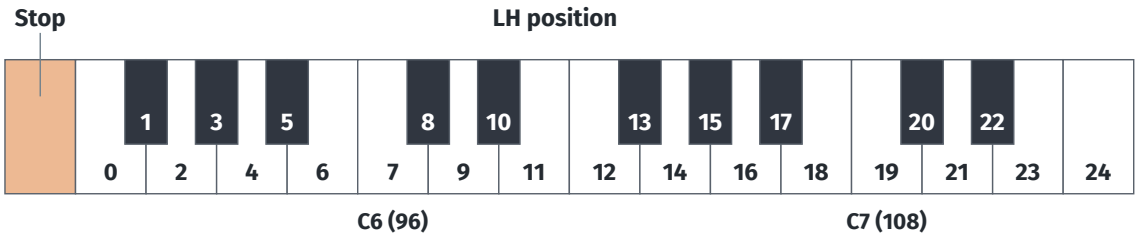
Playable ranges

These ranges assume all six strings are enabled, default tuning, and `Number of frets` set to 24. Bars include both endpoints. Their overlap shows pitches available on more than one string, rather than allocation priority. Retuning or changing the fret count changes these ranges.



Stopping and hand position

Numbers on the keys indicate LH positions, from `0` to 24, on MIDI notes **89–113**. **E5 (88)** triggers `Stop All Notes`.



Style notes use the hold/latch velocity rule; articulation, Stroke, Force String, Handling Noise, and Chord notes are momentary. Force String applies in `Single channel` mode. If `Chord mode input` is mapped to CC, **64–127** enables `Chord Mode` and **0–63** disables it.

Note	Function
13	`Chord mode input`
14	`Handling Noise input`
15	`Tapping style input`
17	`Hammer-on / pull-off input`
18	`Finger style input`
19	`Legato slide input`
20	`Pick style input`
21	`Force A String` (string 4)
22	`Slap/Pop style input`
23	`Force B String` (string 6)
24	`Force C String` (string 1)
25	`Force Index / Slap / Up input` (Stroke 1)
26	`Force D String` (string 3)
27	`Force Middle / Pop / Down input` (Stroke 2)
28	`Force E String` (string 5)
29	`Harmonics input`
30	`Ghost note input`
31	`Force G String` (string 2)
88	`Stop All Notes`
89–113	LH positions 0–24

`LH fret range start` sets the first note of a block reserved for 25 hand positions. Only configured frets respond. The preceding note is always `Stop All Notes`; it takes priority over overlapping assignments.

6.2 Default controllers

The defaults below assign continuous MIDI sources to the instrument controls. If a source is shared by several mappings, each matching control can respond.

Input	Function
Pitch Wheel	Bend up / slide down, split at center.
Channel aftertouch	Fretting pressure.
CC1	`Vibrato depth` scaling.
CC7	`Instrument volume position`.
CC8	`Pickup blend`.
CC12	`Picking hand position`.
CC14	`Fretting hand position`.
CC19	`Vibrato rate`.
CC20	`Pick scrape position`.
CC64	`Let Ring`; **64–127** hold, **0–63** release.
CC67	`Palm damping`.
CC74	`Instrument tone position`.

6.3 Shared mappings and controller updates

Assignable CC numbers are **1–119**; note numbers and values span **0–127**. Shared assignments can affect multiple controls. **CC120–127** retain system/passthrough roles.

The palm, volume, tone, and blend CC mappings update their global controls and respect the `Keyswitch MIDI channel` filter. Manual edits last until the next matching packet. These updates notify the host without recording knob automation; retain the MIDI performance.

6.4 Note pairing and stopping

`Stop All Notes` starts normal damping, cancels attacks/slides, and bypasses `Let Ring` for existing notes. Held switches and controller state remain. `Panic` and `MIDI Reset` immediately silence and reset MIDI pairing. **CC123** releases notes on its channel; **CC120** immediately silences them.

Note-offs pair with note-ons by channel/pitch in arrival order, including velocity-zero note-ons. An older note cannot mute a newer owner of the same string.

7 Sound configuration

The sound of RCModalBass develops from the interaction between its modeled strings and playing gestures. String construction, tuning, and damping determine how the instrument responds, while a gesture's force, position, and release shape the vibration it produces. Pickups convert that vibration into a signal, which the electronics further shape through filtering, circuit loading, or EQ.

7.1 Configuring the strings

The `Instrument` page configures the strings, including their number, fret range, scale length, tuning, construction, sustain, and action.

7.1.1 Basics

`Enabled strings` sets how many strings are available for note allocation, while `Number of frets` bounds their playable positions.

`Fretted / fretless` determines how a stopping position is represented: `Fretted` uses discrete fret positions, while `Fretless` permits fractional positions and uses a different contact model.

`Open scale length` sets the vibrating length of an open string. Changing it also changes the tension needed to maintain the requested tuning.

7.1.2 Tuning

The `Tuning` subpage assigns each string a sounding open MIDI note and a fine detune in cents. `A4 reference` sets the frequency of A4 used to calculate all string tunings.

7.1.3 String construction

The *winding model* uses construction profiles to estimate each string's mass, bending rigidity, and losses. `String N winding` selects the profile, and `String N gauge` sets the string's diameter. Gauge changes both mass and rigidity, while tension is recalculated to preserve the selected tuning.

These properties determine *inharmonicity*: the departure of upper partials from exact harmonic multiples due to bending rigidity. `Reference inharmonicity` provides the shared calibration, and `String N stiffness` multiplies it for each string. The model then accounts for gauge, winding profile, tension, and vibrating length to obtain the effective inharmonicity. A stiffness value of `1` leaves that calibration unchanged; a reference value of `0` removes this rigidity-related contribution.

Construction, tuning, and intrinsic loss settings are captured by new plucks or taps. Changing them prepares the next excitation; a string already ringing keeps its captured configuration.

`Vibration-induced pitch rise` controls extra tension caused by the vibration itself. Stronger vibration can then begin sharper and settle as it decays.

7.1.4 Sustain and tone

RCModalBass gives each *vibration mode* its own decay rate. The note's brightness can therefore change during its decay rather than following a single amplitude envelope. The `Sustain / tone: Low-frequency decay T60` control sets the baseline decay time, or *T60*: the time for amplitude to fall by 60 dB. Additional losses shorten this baseline according to frequency and playing conditions.

`Extra loss at 1 kHz` adds frequency-dependent damping, while `Frequency loss exponent` shapes its distribution across the spectrum. Increasing high-frequency loss makes upper partials disappear sooner. The `String Sustain/Tone` subpage refines this response with `String N High-frequency loss`, `String N sustain`, and `String N age`. The sustain ratio scales the baseline decay time; age contributes a separate frequency-dependent loss.

`Bend motion damping` adds loss during bends. `Fret level taper` and `Fret sustain taper` adjust level and decay as the stopping position rises. Their curves remain referenced to fret 24 even when `Number of frets` is lower.

7.1.5 Action and contact

Action defines the clearance available to the vibrating string. `Open-string action above fret 12 crown` sets the open-string clearance over fret metal, and `String N Action offset above fret 12` adjusts it per string. With less clearance, a smaller vibration can reach the contact surface.

In `Fretted` mode, the string can collide with a higher fret as it vibrates. These collisions produce buzz and clank according to the available clearance and the string's motion. Contact can therefore change during a note as its vibration decays.

In `Fretless` mode, action determines clearance to the modeled fingerboard. Stopped notes can contact that surface, changing their vibration and decay. The board ends one fret interval beyond the last playable position, which bounds the available contact region. Open strings bypass this stopped-note model.

7.2 Shaping the attack

The `Excitation` page defines the right-hand attack models for Finger, Pick, Slap, and Pop. Their parameters describe the displacement, force, contact area, and release that produce each excitation. Tapping uses left-hand excitation; its force settings are described under Fretting.

7.2.1 Velocity response

`Low-velocity attack softness` softens gentle attacks, reducing their high-frequency content so velocity can affect timbre as well as amplitude. The conversion from MIDI velocity to strength is configured under Performance setup.

7.2.2 Finger

Finger excitation begins with a displaced string and releases it. `Pluck displacement at velocity 127` sets how far the string is displaced at maximum velocity. Larger displacements produce stronger attacks. `Finger contact width` spreads the displacement over a finite area, while `Finger release softness` smooths the release and reduces upper harmonics. At large softness values, the attenuation can extend to the fundamental, particularly on higher notes.

`Finger spacing` separates the index and middle attack centers, with middle closer to the bridge. The two fingers share width and softness. Finger harmonics use a sharper release; `Low-velocity attack softness` can soften that release on gentle attacks.

7.2.3 Pick

Pick excitation develops while a moving compliant contact loads the string and then slips off. `Pick contact width` sets its footprint and `Pick depth` sets stroke travel. `Pick stiffness` and `Pick contact loss` shape the loading response, while `Pick edge angle` changes the modeled escape from contact. An angle of `0` represents a square-on edge.

`Pick stroke spacing` separates the up and down attack centers; at `0` they coincide. `Pick attack scratch` adds a short winding rub before loading. Its amount changes scratch level, while stroke speed determines the lead time.

7.2.4 Slap and Pop

Slap applies a short inward force, whereas Pop releases an outward displacement. `Slap strength` changes strike speed and `Slap attack` changes the force duration from thump toward snap. `Slap clank` adjusts the fret reaction. For Pop, `Pop pull` sets displacement, `Pop contact hardness` shapes collisions, and `Pop body` adjusts low-frequency voicing.

Because these gestures can drive the string into contact, changing their strength or pull can change both level and clank. `Slap/Pop spacing` places the Slap center toward the nut and the Pop center toward the bridge. `Slap/Pop compression` reduces the observed peaks without makeup gain. It acts before electronics, so later gain can still raise the output.

7.2.5 Technique levels

The output trims allow techniques to be balanced without changing the energy in the string model. `Pick Level`, `Slap Level`, `Pop Level`, and `Tap / harmonic level` act after pickup observation and before electronics.

The shared Tapping/Harmonics trim includes Pull-off returns in Tapping mode. Harmonics played with Pick receive both it and `Pick Level`. Per-string output trims are described with the electronics below; `Output` on `Play` sets the final instrument level.

7.3 Fretting

The `Fretting` page controls the forces and losses associated with stopping a string, along with the response at fret or fingerboard contacts. Action sets the clearance to a surface; these controls determine the reaction when contact occurs.

7.3.1 Left-hand attacks

Tapping and HO/PO inject a finite finger force using the shared `Finger contact width`. `LH impact amount` sets the added left-hand impact, `LH attack sharpness` shapes its attack, and `Pull-off strength` scales the pull-off contribution.

7.3.2 Damping

`Stopped-end damping` sets the loss at the stopped end of a fretted string. This setting is captured when the note starts. Fretless stopped notes obtain finger loss through their contact model instead.

`Ghost decay T60` sets the baseline decay time captured by a new Ghost Note, while `Ghost-note spatial damping` adds loss according to the overlap between the vibration modes and a finger patch. These losses combine with the string's intrinsic decay.

7.3.3 Fretting pressure

The pressure input adds tension above a normal grip on held, stopped strings. `Full-pressure tension increase` sets how much tension full pressure adds. Increasing it makes pressure raise the pitch farther.

On fretless stopped notes, pressure also slightly firms the finger contact and increases its loss when `Fret / fingerboard contact` is nonzero.

7.3.4 Contact response

`Fret / fingerboard contact` controls coupling to the contact model selected by `Fretted / fretless`. `Contact stiffness` and `Contact damping` shape the reaction when the vibrating string reaches a surface. Setting the coupling amount to `0` bypasses the mechanism.

The fretted model applies a reaction at higher-fret contacts. The fretless model uses compliant finger and board contacts on stopped notes. The resulting sound depends on both these response settings and the clearance described under Action and contact.

7.4 Pickups

The pickups convert modeled string motion into a signal by observing it over specific *sensing regions*. Their geometry determines which modal components contribute to the output. Changing pickup settings updates this observation while preserving the vibration already present in the string.

7.4.1 Position and sensing

`Pickup N distance from bridge` places the sensing region along the string. Each mode has a different spatial shape, so moving the region changes the balance of captured partials. `Pickup N sensing width along string` averages motion across a wider or narrower region, further changing that balance.

`Pickup N pole geometry` selects `Poles` or `Rail`, and `Pickup N field width across string` describes the field's spatial coverage. These controls belong to the sensing model; they do not move the excitation point.

7.4.2 Coils and blend

`Pickup N coil arrangement / wiring` determines how sensing regions combine. The implemented arrangements are `Single`, `Split P-style`, `Dual series`, and `Dual parallel`. `Pickup N coil spacing along string` separates the dual-coil regions, while `Pickup N split-coil stagger` offsets the split regions.

`Pickup N profile` loads a set of geometry, sensitivity, and electrical values. Those controls remain editable; changing a profile-controlled value returns the profile selection to `Custom`.

The enable controls select which pickups contribute, and `Pickup blend` combines their signals in every electronics model. `Pickup 2 polarity` can reverse the second contribution, changing cancellation when both are combined.

7.4.3 Height and response

`Pickup 1 string gap` and `Pickup 2 string distance` set the pickup/string separation used by the height response. A smaller gap increases sensitivity within the model's bounds. `Pickup N sensitivity` scales the pickup output independently of that geometry.

`Pickup N magnetic pull` additionally changes string stiffness through *magnetic loading*. `Pickup nonlinearity amount` instead changes how local motion is converted into the observed signal. At `0`, observation is linear. These are separate controls because magnetic loading affects the vibrating string, while observation nonlinearity affects its electrical representation.

7.5 Electronics

The electronics process the pickup signal. `Electronics model` selects how much of that response is represented: a direct pickup output, a compact filter, or a loaded circuit with optional active EQ.

7.5.1 Choosing a model

Value	Signal path
`Direct`	The observed pickup signal bypasses the electronics filter.
`Simple`	A resonance/filter response is set by `Simple resonance frequency` and `Simple resonance Q`.
`Passive`	The pickups feed a loaded circuit with instrument volume and tone controls.
`Active`	A buffered circuit response feeds preamp gain and three-band EQ.

`Instrument volume position` and `Instrument tone position` operate in `Passive` and `Active`. `Pickup blend` remains available in all four models.

7.5.2 Circuit loading

In the circuit models, the pickups and connected components form a shared electrical network whose *loading* shapes the pickup response. The `Pickup N per-coil resistance`, `Pickup N per-coil inductance`, and `Pickup N per-coil capacitance` controls combine with `Volume-pot resistance`, `Tone-pot resistance`, `Blend-pot resistance`, `Tone capacitor`, `Input load resistance`, and `Cable / load capacitance` to determine the loaded response.

Changing these values can shift the resonance and its damping as well as the action of the instrument controls. They are distinct from the frequency and Q controls used by `Simple`, which describe that model's filter directly.

7.5.3 Active EQ and level balance

The `Active` model adds bass and treble shelves and a peaking mid band. `Bass shelf frequency`, `Mid frequency`, and `Treble shelf frequency` place the bands, while `Bass`, `Mid`, and `Treble` set their boost or cut. `Mid Q` sets the mid band's width: lower values broaden it and higher values narrow it without changing the specified center gain.

`Active EQ voicing` loads band frequencies while retaining the current gains. The frequencies remain editable. `Preamp gain` changes the active-stage level, while `Active Preamp input resistance` and `Active Preamp output resistance` describe its loading.

The `String N Output trim` controls on `Instrument` adjust individual strings after electronics.

7.6 Noise and variation settings

Noise and variation describe two different aspects of a gesture. Noise controls set the audible contribution of contact and movement. Variation changes the parameters captured by successive gestures, which can also change their tone and contact behavior.

7.6.1 Contact and friction

`Fret buzz/clank level` scales the audible contribution produced by metal contact. At −60 dB that contribution is muted, but the mechanical coupling defined under Fretting remains. The fretless board-contact model bypasses metal buzz.

Sliding friction follows a continuous movement trajectory along the wound string. `Slide friction amount` scales it, and `Slide noise texture` changes the balance between scrape and squeak. `Slide noise level speed response` controls how level follows movement speed, while `Slide speed-to-spectrum tracking` controls the associated spectral change.

`Slide texture rate at 0.1 m/s` and `Slide fine texture rate ratio` describe the winding texture; `Slide noise high-pass` and `Slide noise low-pass` bound its spectrum. The texture can remain audible on muted or fretless strings, but ordinary note placement alone does not create the required movement.

`Pick scrape amount` scales explicit pick scraping. Its winding response depends on construction: `Plain` has no winding scrape, while `Flatwound` and `Tapewound` use quieter surfaces. The scratch preceding a normal Pick attack is controlled separately by `Pick attack scratch`.

7.6.2 Variation

Variation lets each accepted excitation capture offsets in position, strength, softness, and contact width. `RH position variation`, `Excitation strength variation`, `Release softness variation`, and `Contact width variation` set the respective ranges. The offsets are sampled for the gesture and do not fluctuate continuously during its sustain.

`Fretting placement error probability` determines how often a placement error occurs, and `Maximum fretting placement error` bounds its size. In fretted mode the error changes the seating/contact approximation; in fretless mode it offsets the stopping point and can detune the note.

`Variation seed` initializes deterministic per-string sequences. Setting it to `0` bypasses variation. A nonzero seed produces the same offsets from the same starting state and accepted event sequence.

8 Sound model and limitations

8.1 Modal string synthesis

RCModalBass represents each string as a sum of vibration modes. Each mode has its own frequency, decay rate, and shape along the string. Their combined motion forms the vibrating string. Construction and tension determine the mode frequencies, while losses determine how the balance of partials changes as a note decays.

For a speaking length L, displacement at position x is represented by

$$y(x,t) = \sum_{n=1}^{M} q_{n(t)} \sin\Big(\frac{n\pi x}{L}\Big).$$

Here x is measured from one end of the speaking span, t is time, M is the number of retained modes, and q_n is the displacement of mode n. The sine term describes that mode's shape along the string.

With fixed coefficients and no further excitation or contact, each mode follows a decaying oscillation:

$$q_{n(t)} = A_n e^{-\alpha_n t} \cos(2\pi f_n t + \varphi_n), \quad \alpha_n = \frac{\ln(1000)}{T_{60,n}}.$$

A_n and φ_n describe the mode's initial amplitude and phase; f_n is its frequency. $T_{60,n}$ is its effective amplitude-decay time in seconds, including the losses acting on that mode. It need not equal the baseline T60 slider value. Contact, excitation, and changing pitch modify this free-decay behavior.

The stiffness-adjusted mode frequencies are calculated as

$$f_n = n f_1 \sqrt{\frac{1 + Bn^2}{1 + B}}.$$

f_1 is the fundamental frequency and B is the effective inharmonicity coefficient derived from the string configuration and speaking length. With $B = 0$, the frequencies are exact multiples of f_1; positive B places the upper modes progressively above those multiples. The denominator keeps the first mode at f_1.

Playing gestures excite these modes in different proportions. The attack position and contact width determine which modes the gesture can drive; its force and release shape their strength. Existing vibration can carry into the next gesture on the same string. Muting and contact with frets or the fingerboard then act on that motion, so their effect depends on how the string is vibrating at that moment.

8.2 From vibration to output

Pickups observe the combined motion at their sensing regions. A mode can contribute strongly at one pickup position and weakly at another, so pickup placement changes the balance of partials in the signal. The electronics process this observed signal before the final output. Handling and friction models supply additional sounds associated with contact and movement.

In the linear pickup model, the observed signal is proportional to a weighted sum of modal velocities:

$$s(t) \propto \sum_{n=1}^{M} W_n \frac{\mathrm{d}q_{n(t)}}{\mathrm{d}t}.$$

W_n combines the sensing geometry, coil wiring, and sensitivity for mode n. Modes that cancel within the sensing regions contribute less to the output. The optional nonlinear pickup response goes beyond this linear sum; the electronics process the resulting signal afterward.

8.3 Approximations

The contact and pickup controls describe approximations used by RCModalBass. Fret collisions use representative sites rather than every fret at once. Fretless contact retains an ideal stopped span with compliant finger and board regions, rather than solving a complete moving endpoint. Pickup fields and magnetic loading are simplified, and the active electronics do not model battery behavior or op-amp clipping.

9 Performance notes

9.1 Real-time performance

`Maximum modes` on `Diagnostics` limits the vibration modes calculated per string. Higher counts can retain more upper partial detail, with modes near the sample-rate limit tapered by the engine.

Processing cost also depends on the number of ringing strings, active contact, and nonlinear pickup observation. The same mode limit can therefore have different real-time costs in different passages. Reducing it or disabling unused model features can lower the load; the host's audio buffer and rendering options provide further ways to manage a demanding project.

9.2 Troubleshooting

- **Note not sounding**: check keyswitch conflicts, `Enabled strings`, `Number of frets`, `Capo Position`, string/channel selection, `Hand span`, `Lock LH Position`, and `Chord Mode`.
- **Quiet high notes**: check `Finger release softness`, damping, and pickup cancellation.
- **No handling sound**: establish movement/selection; hold Handling Noise before Stroke for a landing.
- **Pedal fails to preserve a note**: check channel and string reuse.
- **Control has no effect**: check model context, captured-at-attack behavior, and whether the control is labeled `Not implemented`.

10 Credits and use

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11 Links

Forum thread: https://forum.cockos.com/showthread.php?p=2972988

ReaPack repository: https://github.com/RCJacH/ReaScripts

ReaPack index: https://github.com/RCJacH/ReaScripts/raw/master/index.xml

Demonstration video: https://youtu.be/ZtqFVSdDa5c

12 References

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  - Background on changing pitch and inharmonicity during decay.
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- *Issanchou et al. (2018) — String/frets contacts in the electric bass sound: Simulations and experiments*
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- *Novak et al. (2018) — Experimental Study of Guitar Pickup Nonlinearity*
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- *Giampiccolo, Bernardini & Sarti (2023) — Virtualization of Guitar Pickups through Circuit Inversion*
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- *Robert Bristow-Johnson / W3C — Audio EQ Cookbook*
  - Resonant low-pass, shelving and peaking-EQ coefficients, including Active Mid Q.
- *Fender — What's a Pickup Pole?*
  - Pole geometry, string balance and pickup-height considerations.
- *Seymour Duncan — Humbucker series/parallel wiring diagram*
  - Coil connection topology.
- *Fraunhofer IDMT — IDMT-SMT-Bass dataset*
  - Referenced as recorded articulation comparison material.